

Paper #26b: Sterile Neutrino Mass Generation & UQFF (Redirect Notice)

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Session: Phase 1 (Sessions 1&43)

Framework: UQFF Star-Magic ($\hat{I}^0 = 0.0005/\text{day}$, $[SSq] = 0.57$, $\hat{I}^2_i = 0.61$)

Source: CondensedPhysics2.py, MAIN_1_CoAnQi.cpp

Cross-links: PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md (canonical), PAPER_027 (Lepton Flavor Violation), PAPER_028 (BSM Coupling Constants)

Abstract

This file is a redirect stub. All sterile neutrino mass generation content & including the GUT-scale sector, Yukawa coupling matrix, seesaw mechanism, neutrinoless double beta decay constraints, and UQFF vacuum-mediated mass generation equations & has been merged into the canonical whitepaper PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md. See that file for full derivations and numerical results.

Key parameters consolidated: sterile neutrino mass $M_N \sim 1e14$ GeV, active-sterile mixing $|U_{eN}|^2 < 1e-3$, vacuum contribution $\hat{I}'m_{\hat{I}^{1/2}} = \hat{I}^0 \tilde{A} [SSq] \tilde{A} m_N$.

MERGED INTO: PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md

Key Equations (Summary)

The UQFF-modified seesaw mass formula:

$$m_\nu = m_D^2/M_N + \delta m_{UQFF}$$

$$\delta m_{UQFF} = \kappa \times [SSq] \times \frac{v^2}{M_N} = 5.0 \times 10^{-4} \times 0.57 \times \frac{v^2}{M_N}$$

$$|U_{eN}|^2 < 1.0 \times 10^{-3} \quad (\text{experimental constraint})$$

Numerical summary: $M_N = 1.0e14$ GeV, $m_D = 1.74e2$ GeV (top-quark scale), $\hat{I}'m_{UQFF} \hat{A} 2.85e-4 \tilde{A} v^2/M_N$, $\hat{I}^0[SSq] = 2.85e-4$

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Merged by: GitHub Copilot

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day¹, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments. LHC direct-search results and arXiv constraints bound the parameter space.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\tau}^0$	5.0×10^4 day ¹	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
$\hat{\tau}_i^2$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	$10 \times \hat{A}^2 \hat{A}^2$	Inertia tensor scale
$E_{react}(0)$	$10 \times \hat{A}^2 \hat{A}^2$ J	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{A}^2$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{react} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{A}^2 \hat{A}^2 \hat{A}^2 \hat{A}^2$	4th Advection term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{I}_{\text{c}}=10\hat{\mu}\hat{\text{A}}^1\hat{\mu}^\circ$, $\hat{I}_{\text{a}}=10\hat{\mu}\hat{\text{A}}^1\hat{\text{A}}^2$, $\hat{I}_{\text{b}}=10\hat{\mu}\hat{\text{A}}^1\hat{\text{A}}^1$, $\hat{I}_{\text{d}}=10\hat{\mu}\hat{\text{A}}^1\hat{\text{A}}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{c}}$	$10\hat{\text{A}}^1\hat{\mu}\text{ kg/m}\hat{\text{A}}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\text{d}}$	$2\hat{\mu}/(434\hat{\text{A}}\cdot365.25)\text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_{sum} + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\mu}!$)	Multi-scale field interactions
Buoyant	$\hat{I}_{\text{c}}^2 \hat{\mu} \hat{\text{A}} \text{ Ubi}$	Expanding nebulae, stellar winds
Superconductive	$\hat{\mu} \hat{\text{A}} (1 + 10\hat{\text{A}}^1\hat{\text{A}}^3\hat{\text{A}}\cdot f_{\text{H}})$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.